pyAmpliCol at one loop

Benchmarks of the recursive one-loop engine against MadLoop, Recola and OpenLoops

report 2026-11-07 · newest engine measured `aaff892d` · finite part, binary64, one pinned thread, $B = 128$
Table 1: engine `ca602436`, G1 2026-11-06, AMD EPYC 9754, shared host · Table 2: engine `aaff892d`

×1/16 faster — ×1/4 — ×1 — ×4 — ×16 slower
run time ÷ fastest provider

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **Born-interfered one-loop processes** (the one-loop amplitude interfered with the Born). How to read it: after Table 3.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: engine `ca602436`, native `loop-native-v2-slice-ca602436` (`4cbb549f`…), G1 2026-11-06 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) Rc·OL·ML ≤ 1.3e−12 16 pts · median 1.1e−14 · †	**5.70 s** ×40 0.57 GiB peak · plan 0.0240 MB **Rc 142 ms** · ML 9.70 s	**1.29 ms** ×0.64 $B = 128$ **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms no gate (2→2) numerator 51 %	**0.596 ms** LC/FC 0.46 $B = 128$ · FC 1.29 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 7.9e−13 · Rc 3.3e−10 16 pts · median 4.8e−14 · ¶	**6.73 s** ×17 0.58 GiB peak · plan 0.0300 MB **Rc 406 ms** · ML 9.87 s	**1.72 ms** ×1.1 $B = 128$ **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms <5 ms: not gated ×0.87 numerator 38 %	**1.12 ms** LC/FC 0.65 $B = 128$ · FC 1.72 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 1.2e−11 · Rc 7.2e−11 16 pts · median 1.4e−13 · ¶	**7.07 s** ×15 0.58 GiB peak · plan 0.0360 MB **Rc 462 ms** · ML 10.9 s	**2.64 ms** ×0.90 $B = 128$ **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms <5 ms: not gated ×0.69 numerator 42 %	**1.59 ms** LC/FC 0.60 $B = 128$ · FC 2.64 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) ML·Rc ≤ 7.5e−14 16 pts · median 6.8e−15	**10.1 s** ×14 0.61 GiB peak · plan 0.0530 MB **Rc 701 ms** · ML 14.0 s	**0.788 ms** ×0.83 $B = 128$ **ML 0.948 ms** · Rc 1.08 ms · OL no library <5 ms: not gated ×0.83 numerator 36 %	**0.446 ms** LC/FC 0.57 $B = 128$ · FC 0.788 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 4.9e−14 · ¶	**6.62 s** ×10 0.58 GiB peak · plan 0.0490 MB **Rc 634 ms** · ML 17.6 s	**4.34 ms** ×0.35 $B = 128$ **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms <5 ms: not gated ×0.27 numerator 53 %	**2.38 ms** LC/FC 0.55 $B = 128$ · FC 4.34 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) OL·Rc·ML ≤ 3.7e−7 16 pts · median 5.8e−13 · †	**8.10 s** ×1.2 0.59 GiB peak · plan 0.112 MB **Rc 6.63 s** · ML 47.0 s	**20.1 ms** ×0.21 $B = 128$ **OL 97.0 ms** · ML 247 ms · Rc 351 ms pass ×0.16 numerator 50 %	**7.59 ms** LC/FC 0.38 $B = 128$ · FC 20.1 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc·ML ≤ 2.2e−10 16 pts · median 6.0e−14 · †	**9.53 s** ×11 0.62 GiB peak · plan 0.0680 MB **Rc 833 ms** · ML 24.8 s	**4.21 ms** ×0.41 $B = 128$ **Rc 10.2 ms** · ML 27.3 ms · OL no library <5 ms: not gated ×0.32 numerator 54 %	**1.71 ms** LC/FC 0.41 $B = 128$ · FC 4.21 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 8.8e−9 16 pts · median 1.2e−11 · †	**9.98 s** ×0.16 0.64 GiB peak · plan 0.320 MB **Rc 61.4 s** · ML 218 s	**79.8 ms** ×0.17 $B = 128$ **OL 472 ms** · Rc 1.97 s · ML 5.45 s pass ×0.13 numerator 44 %	**28.2 ms** LC/FC 0.35 $B = 128$ · FC 79.8 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 8.0e−12 · OL 2.3e−10 16 pts · median 1.7e−12 · †	**9.86 s** ×0.30 0.65 GiB peak · plan 0.382 MB **Rc 32.8 s**	**65.3 ms** ×0.17 $B = 128$ **OL 374 ms** · Rc 908 ms · ML not exported pass ×0.13 numerator 57 %	**25.3 ms** LC/FC 0.39 $B = 128$ · FC 65.3 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) ML·Rc·OL ≤ 1.5e−7 8 pts · median 2.5e−11 · †	**14.0 s** ×0.0066 0.89 GiB peak · plan 1.23 MB **Rc 35.4 min** · ML 144.6 min	**307 ms** ×0.11 $B = 128$ **OL 2.84 s** · Rc 17.6 s · ML 38.7 s pass ×0.083 numerator 51 %	**126 ms** LC/FC 0.41 $B = 128$ · FC 307 ms

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — loop · $2 \to 2$ to $2 \to 5$: engine ca602436, native loop-native-v2-slice-ca602436 (4cbb549f…), G1 2026-11-06 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) OL·ML·Rc ≤ 4.5e−11 16 pts · median 5.3e−13	**12.1 s** ×1.8 0.63 GiB peak · plan 0.0960 MB **Rc 6.73 s** · ML 58.6 s	**13.9 ms** ×0.39 **Rc 35.3 ms** · OL 51.0 ms · ML 132 ms $B = 128$ pass ×0.30 numerator 48 %	**5.75 ms** LC/FC 0.41 $B = 128$ · FC 13.9 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) ML·Rc·OL ≤ 4.3e−11 16 pts · median 1.4e−12 · †	**11.1 s** ×0.83 0.64 GiB peak · plan 0.179 MB **Rc 13.4 s** · ML 146 s	**47.9 ms** ×0.19 **Rc 251 ms** · OL 346 ms · ML 1.42 s $B = 128$ pass ×0.15 numerator 63 %	**9.42 ms** LC/FC 0.20 $B = 128$ · FC 47.9 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) OL·Rc·ML ≤ 1.1e−8 16 pts · median 2.7e−11 · †	**11.8 s** ×0.43 0.64 GiB peak · plan 0.176 MB **Rc 27.7 s** · ML 165 s	**51.1 ms** ×0.13 **OL 404 ms** · Rc 620 ms · ML 3.04 s $B = 128$ pass ×0.097 numerator 57 %	**10.1 ms** LC/FC 0.20 $B = 128$ · FC 51.1 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · 1 599 families OL 2.7e−10 · ML 2.8e−9 16 pts · median 4.5e−10	**276 s** ×12 9.24 GiB peak · plan 0.445 MB **Rc 23.5 s** · ML 11.9 min	**131 ms** ×0.13 **Rc 987 ms** · OL 1.40 s · ML 7.07 s $B = 128$ pass ×0.10 numerator 46 % model: the workspace's py3 G_mu copy	***not supported*** refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage)
	$u\bar{u} \to ggggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 2.1e−12 · Rc 2.9e−10 3 pts · median 7.0e−11 · ¶	**88.2 s** ×0.0086 2.76 GiB peak · plan 14.7 MB **Rc 170.3 min** · ML 12.2 h	**1.31 s** ≥×0.0072 **Rc ≤182 s** · ML 569 s · OL no library $B = 128$ pass ×0.0023 numerator 62 %	**438 ms** LC/FC 0.34 $B = 128$ · FC 1.31 s
	$gg \to ggggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 1.5e−7 10 pts · median 3.4e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**133 s** 4.39 GiB peak · plan 20.2 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**4.99 s** $B = 128$ no gate (no matched provider) numerator 52 %	**2.68 s** LC/FC 0.54 $B = 128$ · FC 4.99 s
	$2 \to 6$: pyAmpliCol only, no provider exists at $2 \to 6$, so no ratio, no gate and no colour · engine 2f3f76d9 (work/v2-mf-8g-3), native loop-native-mf-2to6-2f3f76d9-lto (2bae17f9…), an ancestor of the engine above, not re-run on it · shared host, indicative · internal checks only			
$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.4e−3 (2 pts above 10^{-5}) · binary64 = DD to 7.4e−5 · rescue 11/16 · NGluon DD to 6.3e−13 · **FC** ε^{-2} ≤ 8.4e−11 · E ≤ 5.1e−3 (2 pts above 10^{-5}) · Catani ε^{-1} ≤ 7.1e−5 · binary64 = DD to 1.2e−4 · rescue 11/16 Ward FC not run, Ward LC not asserted	**55.3 min** FC 74.0 GiB peak · plan 486 MB · store 25.8 min artifact 0.53 GB · load 4.21 s (1.06 GiB) **LC** 46.1 min · 63.9 GiB · artifact 0.364 GB	**201 s** 1 block of 16 $B = 16$ no provider: no gate inputs 55 % peak 40.1 GiB	**98.0 s** 1 block of 128 $B = 128$ numerator 54 % peak 27.2 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 6.2e−7 · binary64 = DD to 8.4e−11 · rescue 1/16 · Ward ≤ 6.5e−12 · **FC** ε^{-2} ≤ 1.1e−11 · E ≤ 5.9e−6 · Catani ε^{-1} ≤ 3.3e−8 · binary64 = DD to 1.7e−10 · rescue 2/16 · Ward ≤ 2.1e−11	**73.6 min** FC 92.9 GiB peak · plan 814 MB · store 449 s artifact 0.143 GB · load 3.85 s (1.66 GiB) **LC** 20.5 min · 47.0 GiB · artifact 0.047 GB	**62.1 s** 1 block of 128 $B = 128$ no provider: no gate numerator 77 % peak 31.7 GiB	**10.9 s** 1 block of 128 $B = 128$ numerator 60 % peak 12.6 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 40 454 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · E ≤ 7.5e−7 · binary64 = DD to 1.3e−11 · rescue 2/16 · Ward ≤ 4.3e−12 · **FC** ε^{-2} ≤ 2.0e−13 · E ≤ 6.5e−6 · Catani ε^{-1} ≤ 6.6e−11 · binary64 = DD to 1.3e−11 · rescue 4/16 · Ward ≤ 8.5e−12 (DD 1.1e−28)	**29.8 min** FC 42.5 GiB peak · plan 315 MB · store 46.4 s artifact 0.038 GB · load 1.51 s (0.68 GiB) **LC** 506 s · 18.2 GiB · artifact 0.009 GB	**19.2 s** 1 block of 128 $B = 128$ no provider: no gate numerator 75 % peak 36.0 GiB	**2.67 s** 1 block of 128 $B = 128$ numerator 55 % peak 7.38 GiB

Table 2: **Loop-induced processes** ($|A_{\text{loop}}|^2$; the equivalent of Table 1). How to read it: after Table 3.

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, default backend
pyAmpliCol: engine aaff892d, native loop-native-v2-slice-aaff892d (5bfdbad8…) on itphlies (packages landed: lit-amp) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-07 · nothing estimated			
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend
$2 \to 1 \cdot 1 \to 2$	$gg \to h$ 3 legs · QCD · LI-A1 validation pending (lit-val)	**8.88 s** incl. probes 4.71 s (53 %) 0.56 GiB peak · plan 0.0100 MB	**0.290 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→1) **NinjaRS per amplitude**
	$h \to gg$ 3 legs · QCD · LI-A2 validation pending (lit-val)	**8.79 s** incl. probes 4.67 s (53 %) 0.56 GiB peak · plan 0.0100 MB	**0.311 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude**
	$h \to aa$ 3 legs · QED · LI-A3, LI-E1 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet	
	$h \to za$ 3 legs · QED · LI-A4, LI-E1 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet	
	$aa \to h$ 3 legs · QED · LI-E2 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	

pyAmpliCol: engine aaff892d, native loop-native-v2-slice-aaff892d (5bfdbad8…) on itphlies (packages landed: lit-amp) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-07 · nothing estimated

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, default backend
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend
$2 \to 2$	$gg \to aa$ 4 legs · QCD · LI-A5 validation pending (lit-val)	**16.3 s** incl. probes 10.4 s (64 %) 0.70 GiB peak · plan 0.0210 MB	**5.46 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude**
	$gg \to zz$ 4 legs · QCD · LI-21 validation pending (lit-val)	**20.7 s** incl. probes 14.1 s (68 %) 0.78 GiB peak · plan 0.0280 MB	**26.9 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude**
	$gg \to w^+w^-$ 4 legs · QCD · LI-22 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$gg \to za$ 4 legs · QCD · LI-23 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$gg \to hh$ 4 legs · QCD · LI-24, LI-A7 validation pending (lit-val)	**11.9 s** **×140** incl. probes 7.42 s (62 %) 0.60 GiB peak · plan 0.0100 MB Rc 85.1 ms	**0.779 ms** **×1.1** **ML 0.692 ms** · Rc 1.36 ms · OL 1.56 ms $B = 128$ no gate (2→2) **NinjaRS per amplitude**
	$gg \to hz$ 4 legs · QCD · LI-25 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$gg \to hg$ 4 legs · QCD · LI-26, LI-A7 validation pending (lit-val)	**11.5 s** incl. probes 6.76 s (59 %) 0.61 GiB peak · plan 0.0110 MB	**2.39 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude**
	$u\bar{u} \to hg$ 4 legs · QCD · LI-26, LI-A7 validation pending (lit-val)	**9.91 s** incl. probes 5.14 s (52 %) 0.57 GiB peak · plan 0.0100 MB	**0.374 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude**
	$ug \to hu$ 4 legs · QCD · LI-26 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$gg \to zg$ 4 legs · QCD · LI-27 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$gg \to ag$ 4 legs · QCD · LI-27 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) ML · OL · Rc: no provider timing record of the row in li_campaign.json	
	$e^+e^- \to ha$ 4 legs · QED · LI-A4, LI-E3 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	
	$aa \to aa$ 4 legs · QED · LI-A6, LI-E2 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	
	$aa \to zz$ 4 legs · QED · LI-E2 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	
	$aa \to hh$ 4 legs · QED · LI-E2 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	
	$e^+e^- \to hh$ 4 legs · QED · LI-E3 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet	
$1 \to 3$	$h \to ggg$ 4 legs · QCD · LI-37 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet	
	$z \to ggg$ 4 legs · QCD · LI-37 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet	

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, default backend
pyAmpliCol: engine aaff892d, native loop-native-v2-slice-aaff892d (5bfdbad8…) on itphlies (packages landed: lit-amp) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-07 · nothing estimated			
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend
$2 \to 3$	$e^+e^- \to ggg$ 5 legs · QED · LI-E4 validation pending (lit-val)	***pending*** served by lit-ew (not landed at this head) **Rc 4.44 ms** · ML 15.5 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b))	
	$gg \to hhh$ 5 legs · QCD · LI-31 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 5.64 ms** · ML 9.03 ms · OL 22.9 ms	
	$gg \to hgg$ 5 legs · QCD · LI-32, LI-A7 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 25.5 ms** · OL 25.7 ms · ML 86.2 ms	
	$gg \to zzg$ 5 legs · QCD · LI-33 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 178 ms** · OL 183 ms · ML 721 ms	
	$gg \to aag$ 5 legs · QCD · LI-33 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 47.7 ms** · OL 52.5 ms · ML 200 ms	
	$gg \to w^+w^-g$ 5 legs · QCD · LI-33 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 68.8 ms** · OL 84.4 ms · ML 347 ms	
	$gg \to zzz$ 5 legs · QCD · LI-34 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 134 ms** · OL 151 ms · ML 560 ms	
	$gg \to hzg$ 5 legs · QCD · LI-35 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 19.1 ms** · OL 32.3 ms · ML 73.9 ms	
	$gg \to agg$ 5 legs · QCD · LI-35 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 75.8 ms** · OL 78.8 ms · ML 250 ms	
	$gg \to zgg$ 5 legs · QCD · LI-35 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 116 ms** · ML 476 ms · OL unavailable: no OpenLoops ls library serves g g > z g g: the public channel database of OpenLoops 2.1.5 (proclib/channels_public.rinfo, 2026-11-06) has zgg/zggg (ppvj2) but no zgggg channel; row ML/RC only (PLAN §3.2)	
	$u\bar{u} \to hgg$ 5 legs · QCD · LI-36 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 1.53 ms** · OL 2.61 ms · ML 2.63 ms	
	$u\bar{u} \to d\bar{d}h$ 5 legs · QCD · LI-36 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **ML 0.134 ms** · OL 0.271 ms · Rc 0.536 ms	
$2 \to 4$	$gg \to hhgg$ 6 legs · QCD · LI-41 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 189 ms** · OL 572 ms · ML 671 ms	
	$gg \to hggg$ 6 legs · QCD · LI-42 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 577 ms** · OL 853 ms · ML 1.98 s	
	$gg \to e^+e^-\mu^+\mu^-$ 6 legs · QCD · LI-43 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 3.70 ms** · OL 6.64 ms · ML 33.8 ms	
	$gg \to aagg$ 6 legs · QCD · LI-44 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 1.67 s** · OL 1.96 s · ML 5.89 s	
	$gg \to zzzz$ 6 legs · QCD · LI-44 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 5.07 s** · ML 19.9 s · OL unavailable: no OpenLoops ls library serves g g > z z z z (public channel database of OpenLoops 2.1.5, 2026-11-06); row ML/RC only (PLAN §3.2)	
$2 \to 5$	$gg \to e^+e^-\mu^+\mu^-g$ or $gg \to hgggg$ 7 legs · QCD · LI-51 validation pending (lit-val)	***pending*** served by lit-r2 (not landed at this head) **Rc 63.4 ms** · OL 252 ms · ML 1.20 s	
EFT	rank > N+2 EFT row (HC_NLO_X0 or SMEFT) EFT · LI-X1 · TIRust only validation pending (lit-val)	***pending*** served by lit-eft (not landed at this head) ML: no provider timing record of the row in li_campaign.json	

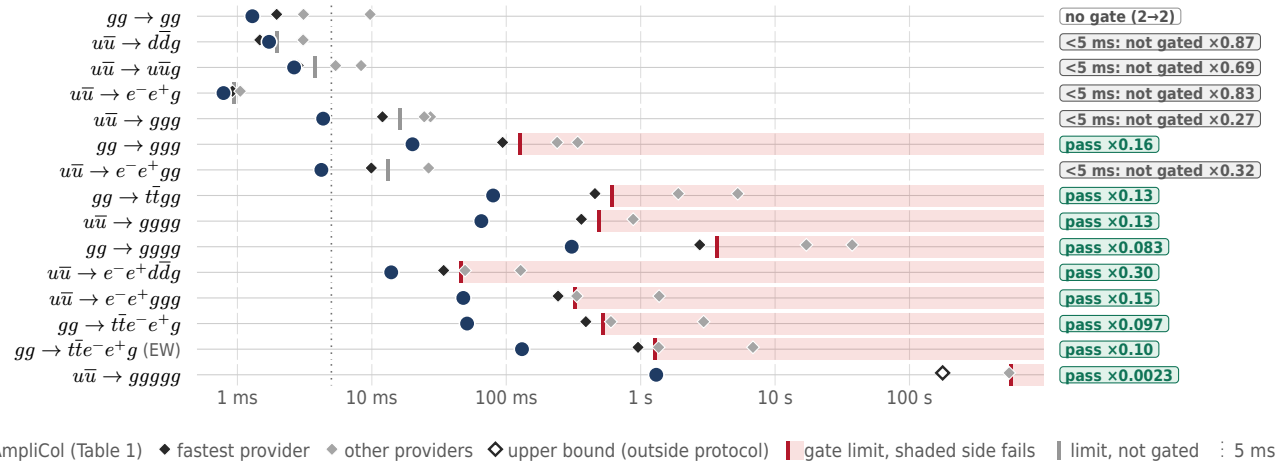


Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider).

Table 3: **Gate and stage split at $B = 128$**, ms per point: the numbers behind the gate chips and stage bars of Table 1 and behind Figure 1. pyAmpliCol's median (full colour), the binding limit and its provider, the uncertainty band of time ÷ limit, the gate chip of Table 1, and pyAmpliCol's time per stage.

Process	ms/pt	Limit	Band	Gate	inputs	numerator	scatter	NinjaRS	masters	glue
$gg \to gg$	**1.29**	—	—	no gate (2→2)	0.191	0.665	0.0663	0.113	0.0372	0.219
$u\bar{u} \to d\bar{d}g$	**1.72**	1.97 OL	0.862–0.884	<5 ms: not gated ×0.87	0.0768	0.647	0.397	0.142	0.109	0.342
$u\bar{u} \to u\bar{u}g$	**2.64**	3.80 OL	0.678–0.711	<5 ms: not gated ×0.69	0.126	1.10	0.424	0.254	0.190	0.525
$u\bar{u} \to e^-e^+g$	**0.788**	0.948 ML	0.830–0.832	<5 ms: not gated ×0.83	0.0516	0.285	0.129	0.0355	0.0262	0.263
$u\bar{u} \to ggg$	**4.34**	16.1 OL	0.259–0.280	<5 ms: not gated ×0.27	0.389	2.32	0.411	0.456	0.211	0.576
$gg \to ggg$	**20.1**	126 OL	0.155–0.164	pass ×0.16	5.57	9.88	0.518	2.27	0.524	1.14
$u\bar{u} \to e^-e^+gg$	**4.21**	13.3 Rc	0.314–0.319	<5 ms: not gated ×0.32	0.161	2.26	0.589	0.435	0.237	0.524
$gg \to t\bar{t}gg$	**79.8**	613 OL	0.128–0.133	pass ×0.13	11.0	34.9	0.888	18.9	6.63	7.49
$u\bar{u} \to gggg$	**65.3**	486 OL	0.130–0.139	pass ×0.13	9.37	37.2	1.74	9.04	2.98	4.78
$gg \to gggg$	**307**	3 690 OL	0.0824–0.0838	pass ×0.083	93.7	158	2.33	38.4	6.05	8.29
$u\bar{u} \to e^-e^+d\bar{d}g$	**13.9**	45.9 Rc	0.278–0.329	pass ×0.30	0.570	6.62	1.79	1.90	1.27	1.58
$u\bar{u} \to e^-e^+ggg$	**47.9**	326 Rc	0.135–0.159	pass ×0.15	1.56	30.1	2.05	7.93	2.75	3.53
$gg \to t\bar{t}e^-e^+g$	**51.1**	525 OL	0.0897–0.105	pass ×0.097	2.88	29.0	0.734	10.2	4.26	4.07
$gg \to t\bar{t}e^-e^+g$ (EW)	**131**	1 280 Rc	0.0935–0.111	pass ×0.10	6.29	60.7	1.75	29.7	16.5	16.2
$u\bar{u} \to ggggg$	**1 310**	569 000 ML	0.00210–0.00248	pass ×0.0023	215	804	9.27	186	46.0	46.8
$gg \to ggggg$	**4 990**	—	—	no gate (no matched provider)	1 970	2 580	11.0	344	42.0	28.4

Reading the tables

Table 1 (Born-interfered one-loop processes) has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to ggggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation multipliers are uncoloured (generation time is not the main metric) and leading colour (LC) has no provider ratio. Grey cells say why a value is missing. Run times are medians per phase-space point at $B = 128$, finite part, binary64, one pinned thread, stability checks off; generation is generate + plan export (model loading excluded) with its peak memory and plan size. The providers run in production mode: MadLoop 3.7.0 (ML), Recola 2.2.5 with COLLIER (Rc), OpenLoops 2.1.5 from prebuilt libraries (OL: no generation time). Each cell shows the most recent measured configuration (`scripts/build_summary.py`, by git ancestry of the revisions), named once in the table's first row (engine, native build and digest, date, host); a cell from another configuration names its origin.

Gate chips (Table 1, Figure 1, Table 3): the time over the binding limit, the smallest of MadLoop's time and 1.30 × each of OpenLoops' and Recola's; **pass** or **fail** when the whole uncertainty band (Table 3) lies below or above 1, else **open**. A process below 5 ms per point has no blocking gate ("<5 ms: not gated", user rule of 2026-10-24) and its chip keeps the measured ratio; $2 \to 2$ has no gate. The gate is taken at $B = 128$ only.

Stage bars (the full-colour cell names the largest share; Table 3 gives every stage in ms): inputs = trees, Born and counterterm inputs; numerator = seeds, fused factors and the dressing of the open-loop objects; scatter = closures into NinjaRS lanes; NinjaRS = the reduction without the masters; masters = the master integrals (canonical arguments, cache, oneloopmaster); glue = point parameters, input conversion, assembly and the rest of the call.

Validation (process cell): the largest deviation $|a-b|\ /\max(|a|,|b|,|B|\ \alpha_s/2\pi)$ of the binary64 finite part per reference at the timed points (NJet DD: NJet 3.1.2 in double-double), and the median over the points and references; † the largest lies within pyAmpliCol's binary64 stability estimate at that point; ¶ it is the reference's own: pyAmpliCol's double-double value lies as far from it, and its binary64 value lies within its estimate of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (1.5×10^{-7} and 2.6×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): binary64 within the estimate, DD = NJet** (estimates E 1.5×10^{-6} and 3.4×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points, passes.

Notes under a value qualify it and leave it as measured, in amber a known limitation, in grey its run conditions: $gg \to t\bar{t}e^-e^+g$, full colour ("model: the workspace's py3 G_mu copy"): timed on reference/models/loop_qcd_qed_sm_Gmu_py3 (the harness's processes.json default, as at M-SM), not on the shipped patched UFO (UFOmodeldb 6ca4b120 + the user's issue-1 patch) that the M-SM campaign validated; the speed gate is taken on this timing.

The $2 \to 6$ rows (`summary.json` `preliminary`, from `data/2to6.json`) have no provider, so no multiplier, gate or colour; their engine is stated once in the table. Generation = generate + plan export, with the store, artifact and load beside it ("FC" marks the full-colour generation, LC below it); the timed evaluation is binary64, finite part, checks off, one pinned thread, at the largest batch measured. The process cell lists the internal checks (ε^{-2}: double pole against the universal value; E: the stability estimate, with the points above 10^{-5}; rescue: the stream points the default rescue policy's binary64 level sends up; Ward on the external gluons; binary64 against double-double; Catani at full colour; NGluon at leading colour on the all-gluon row) and, in grey, those not run or not asserted.

Table 2 follows Table 1′s rules for loop-induced processes, $|A_{\text{loop}}|^2$ without a Born: one row per process of the M-LIT validation matrix (`m-lit/PLAN.md` §4.1; its ids in the process cell), grouped by multiplicity; the backend-unit, zero, identity, observable and complex-mass-scheme rows (LI-B1, LI-B2, LI-Z1, LI-Z1b, LI-Z2, LI-W1, LI-L1, LI-P1, LI-C1, LI-S1, LI-O1) are validation-only and not listed. The cell is the batched $B = 128$ time per point (median, finite part, one pinned thread, checks off) on the default backend TIRust; NinjaRS per amplitude, labelled in the cell, where TIR does not serve the row yet; the other backend's time beside it when both ran. Its multiplier against the fastest matched provider (MadLoop at its loop-induced production setting 7|6|1, OpenLoops `ls`, Recola; production mode) colours it with Table 1′s scale. Generation, uncoloured, is the full production `generate` cost: generate + plan export + the generation-time double-double finiteness and Ward probes, which run for every loop-induced process (their time and share beneath the value), with the peak memory over all three; the provenance line names pyAmpliCol's host and the providers'. Rows of four legs or fewer ($2 \to 1$, $1 \to 2$, $2 \to 2$, $1 \to 3$) are reported without a gate; from $2 \to 3$ upward the gate is Table 1′s (faster than MadLoop and at most 1.30× each matched OpenLoops and Recola time, on TIRust), "<5 ms: not gated" below 5 ms per point, no gate without a matched provider; verdict chips come with the gate tool's records, "no verdict yet" before them ("no provider time yet" while no provider of the row is timed). Grey cells are explicit markers: **pending** (the package that serves the row, named in the cell, has not landed at this head, or the input file that fills the cell does not exist yet), **not run** (a landed package without a record of the row), and a provider's reason where it has no matched time. Nothing is estimated or extrapolated; each cell shows the newest record of its row. At this head 7 of 42 rows are served; inputs: `data/li-timing.json` present, `tools/loop_reference/li_campaign.json` `(provider_timing_records)` present, `data/lit.json` not written yet (lit-val); the registry against `lit-ref`'s manifest (`tools/loop_reference/li_campaign.json`): checked (39 rows, 42 ids).

Figure 1 places each row's batched full-colour value of Table 1 (dots) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider.

Table 3 lists, for every row of Table 1 timed at $B = 128$, the numbers its gate chip and stage bar summarise: the binding limit and its provider, the uncertainty band of time ÷ limit (the standard error of pyAmpliCol's median over its timing blocks plus the provider's run-to-run half-range), and the time of each stage. Data: `data/summary.json` and `data/summary-li.json`, written by `scripts/build_summary.py` from the campaign records in `data/` (`README.md`).